ANSI/IES LM-82-20

Illuminating
ENGINEERING SOCIETY

APPROVED METHOD: CHARACTERIZATION OF OPTICAL AND ELECTRICAL PROPERTIES OF SOLID-STATE LIGHTING PRODUCTS AS A FUNCTION OF TEMPERATURE

AN AMERICAN NATIONAL STANDARD

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Prepared by
IES Testing Procedures Committee

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1.0 Introduction and Scope

1.1 Introduction

The performance (e.g., luminous flux, life) of light emitting diodes (LEDs) depends strongly on the temperature at the LED junction, and this temperature can vary depending on how the LED is integrated into the luminaire and on the application environment. LED light engines and integrated LED lamps are used in many different types of luminaires, including those for decorative lighting and non-directional applications.

Generally, LED performance, in terms of luminous flux, luminous efficacy, color, and life, is affected by the temperature at the LED junction. During operation, the LED junction temperature can reach as high as 150 °C. At such temperatures, the amount of thermal energy transferred to the ambient environment via radiation is very small. Therefore, conductive and convective thermal energy transfer techniques are employed to limit the LED junction temperature within the upper limit for the luminaire design. Active or passive heat sinks or combinations of these are typically used to effectively control the LED junction temperature to the luminaire design value.

When an LED light engine or integrated LED lamp is used in a luminaire, the thermal environment near the LED is altered by both the luminaire design and the application environment. By measuring the performance characteristics of an LED light engine or integrated LED lamp at various temperatures, the luminaire manufacturer can model the expected light output of a given luminaire design by measuring the operating temperature of the LED light engine or integrated LED lamp in the luminaire. For luminaires that cannot be measured as a complete system, modeling the light source portion for light output temperature dependence shall be an option.

In addition, for an LED luminaire that may be installed in a different temperature environment, its optical property will be changed due to the temperature dependence. The comparison of such change can be measured using this document.

1.2 Scope

The purpose of this document is to establish consistent methods of measurement and data presentation for ease of interpretation and comparison, which will assist luminaire manufacturers in selecting suitable LED light engines and integrated LED lamps for each luminaire product. This approved laboratory method defines the procedures to measure optical and electrical properties as a function of temperature of LED light engines and integrated LED lamps. This document is also applicable to LED luminaires.

2.0 Normative References

2.1 ANSI/IES LM-79-19

Illuminating Engineering Society. Approved Method: Optical and Electrical Measurements of LED Lighting Products. New York: IES; 2019.

2.2 ANSI/IES LS-1-20

Illuminating Engineering Society. Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020. Online: www.ies.org/standards/definitions/. (Accessed 2019 May 3).

3.0 Definitions

3.1 device under test (DUT)

For the purpose of this document: a DUT is an LED light engine, or an LED lamp, or an LED luminaire (each as defined in ANSI/IES LS-1-20).

3.2 heat sink

A device attached to an LED assembly (package, array, or module) to dissipate heat.

3.3 thermal chamber

A chamber used to maintain a local ambient temperature.

4.0 Ambient and Physical Test Conditions

4.1 General

This approved method characterizes the DUT performance (e.g., lumen output) as a function of the temperature at manufacturer-specified temperature monitoring points. These performance characteristics can then be estimated over the range of temperatures set by the maximum and minimum temperatures at which the DUT only was tested using a simple curve-fit to the measured values. Using this information, the performance of the DUT in a luminaire can be estimated by measuring the temperature of the manufacturer-specified temperature monitoring points while the DUT is operated in the luminaire at specified operation (e.g., in-situ) conditions and by comparing this to the performance curve derived from the DUT-only measurements.

DUT driver electronics may be affected by the operating temperature. Proximity of the driver electronics to the heat-generating LEDs can result in higher driver component temperatures than are assignable due to general ambient temperature. Manufacturers of LED drivers may specify a maximum operating temperature and identify one or more temperature monitoring points. The temperature of the driver temperature monitoring point (TMP) is correlated with the temperature of the most temperature-sensitive component of the driver electronics. The driver TMP(s) when provided shall be monitored during testing.

For the purpose of this document, a DUT manufacturer-specified TMP temperature is denoted as T_b, and a driver manufacturer-specified TMP temperature is denoted as T_d.

All components of the DUT shall be subject to the same environmental conditions, even though the elements of the assembly may not be mechanically connected (e.g., the driver, though electrically connected, is mechanically separated from LED modules).

4.2 Temperature Monitoring Point and Temperature Measurement Device

The temperature monitoring points shall be identified by the requestor or the DUT manufacturer. The requester shall identify a temperature monitoring point T_b and a temperature monitoring point T_d (if applicable), and shall diagram the temperature monitoring point for each.

A variety of temperature measuring devices may be used such as thermocouples or thermistors. Thermistors (temperature sensitive resistors) may be used and shall be calibrated against a known standard.

The temperature measuring device shall be chosen such that it does not conduct a significant amount of thermal energy away from the DUT. The temperature measurement device shall have a calibration with an expanded uncertainty (k = 2) of less than 1 °C. The temperature measuring device(s) shall be thermally and mechanically attached to the requester or manufacturer-specific test point(s) throughout the duration of the tests.

4.3 Seasoning

For the purposes of characterizing the DUT with respect to temperature, the DUT does not require seasoning.

4.4 Thermal Environment

There are several methods for controlling the temperature of the DUT. The difficulty is integrating the photometric measurement of the DUT while controlling its temperature. One such method is placing the DUT into a temperature controlled integrating sphere. Additional methods include: a) mounting the DUT to a thermoelectric cooler (TEC), or b) mounting the DUT in a temperature chamber that only controls the local environment around the DUT. The most important aspect is to ensure that the method of controlling the DUT temperature has a heating equivalence similar to the application of the DUT. For example, if heat is applied at one end of the heat sink with a thermoelectric cooler, this may be equivalent to a temperature chamber using convection to control the DUT temperature.

Regardless of the temperature controlling method, the T_b of the DUT operating at the electrical conditions specified for the test shall be set to the specified temperature within a tolerance of ±2 °C. For the initial photometric measurements (see **Section 6.1**) the ambient environment shall follow the procedures in ANSI/IES LM-79-19 (see **Section 2.1**).

Photometric detector(s) shall remain within their temperature tolerance range throughout planned testing, as detector thermal variation can introduce increased uncertainties.

4.5 Stabilization

Before all photometric measurements are taken at any given temperature, the DUT shall be operated long enough to reach stabilization and temperature equilibrium. The time required for stabilization depends on the DUT. Stability shall be achieved when the variation (maximum to minimum) of at least three readings of the light output and electrical power consumption—taken at maximum intervals of 10 minutes over a period of 20 minutes and divided by the chronological last of these measurements—is less than 0.5%. The readings should be taken at regular intervals.

5.0 Electrical Test Conditions

5.1 Electrical Settings

The DUT shall be operated at the rated voltage (AC or DC) according to the manufacturer's specification for normal use of the DUT. If the DUT has dimming capability, measurements shall be performed at the power condition for maximum light output. If the DUT has multiple modes of operation, including variable CCT or color tunable, measurements may be made for different modes of operation if necessary or desired, and such setting conditions shall be clearly documented in the test report.

5.2 Circuits

5.2.1 DC. For DUTs requiring DC input power, a combination of 1) a DC digital voltmeter (DVM) or multimeter (DMM) in voltage mode, and 2) a DC digital ammeter (DMA) or digital multimeter (DMM) in current mode shall be used. The DVM shall be connected across the electrical inputs of the DUT; the DMA or DVM (or DMM in current mode) shall be connected inline in the return lead of the circuit. The reported wattage shall be the product of the measured current and voltage. The voltage, current, and wattage shall be documented in the test report.

5.2.2 AC. For DUTs requiring AC input power, an AC digital power analyzer shall be connected between the AC power supply and the DUT. The voltage sensing leads shall be connected across the electrical power inputs of the DUT. Input power, input voltage, and current shall be measured and reported.

5.3 Electrical Measurement Uncertainties

The uncertainties for DC voltage and DC current measurement shall be ≤0.2%.

The expanded uncertainties for AC voltage and AC current measurements shall be ≤0.2% for the range used. The expanded uncertainty for AC power measurement shall be ≤0.25%. In addition, AC measurement instruments shall measure accurately up to the 50th harmonic of the AC fundamental frequency.

6.0 Test Method and Procedures

6.1 Ambient Temperature Measurement Using Absolute Photometry Measurement Methods

6.1.1 Ambient Temperature (Initial) Measurement. The DUT shall be first tested for electrical and photometric characteristics in accordance with the testing methods specified in ANSI/IES LM-79-19 (see **Section 2.1**). Electrical power P_i (W), and the total luminous flux Φ_i (lm), total radiant flux, or total photon flux shall be recorded. Optionally, chromaticity coordinates, (x_i, y_i) or (u'_i, v'_i); correlated color temperature, CCT_i (K); and/or spectral power distribution (SPD) may be recorded. The DUT manufacturer-specified temperature monitoring point temperature(s), T_b as $T_{b,i}$, and the manufacturer-specified temperature monitoring point(s) for the driver, T_d as $T_{d,i}$, shall be recorded.

6.1.2 Ambient Temperature (Calibration) Measurement. As the second step, the DUT shall be measured with a device that controls the temperature(s) ($T_{b,i}$) of the DUT. The temperature controlling method (**Section 4.4**) shall be used to adjust the thermal environment so that T_b is the same as the initial ambient temperature measurement of **Section 6.1**. The value of T_b shall be recorded as $T_{b,0}$. While keeping T_b constant, the electrical power P_0, the total luminous flux Φ_0 or luminous intensity I_0 at the defined spatial point, or the total radiant flux or total photon flux shall be measured. Optionally, chromaticity coordinates (x_0, y_0) or (u'_0, v'_0); correlated color temperature, CCT_0; and/or SPD may be recorded if they were recorded during the initial measurement.

The correction factors between the initial ambient temperature measurements and the room temperature calibration measurements shall be determined as:

Electric power correction factor:

$$C_{\text{power}} = \frac{P_i}{P_0} \quad \textbf{(6-1)}$$

Luminous flux correction factor:

$$C_{\text{flux}} = \frac{\Phi_i}{\Phi_0} \quad \textbf{(6-2a)}$$

or

$$C_{\text{flux}} = \frac{\Phi_i}{I_0} \quad \textbf{(6-2b)}$$

Note: The radiant flux and photon flux correction factors are the same.

Chromaticity corrections:

$$\Delta x = x_i - x_0 \text{ , } \Delta y = y_i - y_0 \quad \textbf{(6-3a)}$$

Or

$$\Delta x = x_i - x_0 \text{ , } \Delta y = y_i - y_0 \quad \textbf{(6-3b)}$$

Color corrections:

$$\Delta \text{ CCT} = CCT_i - CCT_0 \quad \textbf{(6-4a)}$$

$$\Delta\ CRI = CRI_i - CRI_0 \quad \textbf{(6-4b)}$$

$$\Delta\ R_9 = R_{9,i} - R_{9,0} \quad \textbf{(6-4c)}$$

Note: There is no SPD correction.

6.2 Measurement with Temperature Controlled Device

When the DUT is measured with a device that controls the temperature T_b of the DUT (see **Section 4.4**), the photometry measurement may be made either as flux integration or a directional measurement.

- **Flux integration measurement.** A flux integrating device of any geometry of sufficient size to contain the DUT with a method to control the critical temperature values per **Section 4.4** may be used. Acceptable realizations include but are not limited to: a) a temperature controlling method coupled with an integrating sphere using a 2π or 4π configuration; and b) a temperature controlling method coupled with a pseudo-integrating device such as a tube lined with white reflecting material and a cosine corrected spectrometer head mounted on the bottom.
- **Directional measurement.** The measurement can be made at a single spatial point defined and selected by the initial measurement of luminous intensity, I (cd).

Note: For a narrow-beam DUT, the sensitivity of the angular beam properties to the change of temperature should be investigated.

In either of the above measurements, once the DUT with the temperature controlling method is coupled to the measurement system, they shall not be separated until the measurements are complete.

6.2.1 Measurement at Temperature $T_{b,0}$ + 25 °C. The temperature controlling method shall be adjusted so that T_b reaches no lower than $T_b = T_{b,0}$ + 25 °C. This value shall be recorded as $T_{b,1}$. The electrical power P_1, photometric and colorimetric properties, luminous intensity I_1, and/or the total luminous flux Φ_1, radiant flux or photon flux shall be measured. Optionally, chromaticity coordinates, (x_1, y_1) or (u'_1, v'_1); correlated color temperature, CCT_1; and/or SPD shall be measured.

The corrected results for the first elevated temperature, $T_b = T_{b,1}$, shall be recorded as:

$$P = C_{\text{power}} P_1 \quad \textbf{(6-5)}$$

$$\Phi = C_{\text{flux}} \Phi_1 \quad \textbf{(6-6a)}$$

or

$$\Phi = C_{\text{flux}} I_1 \quad \textbf{(6-6b)}$$

And optionally,

$$x = x_1 + \Delta x \text{ , } y = y_1 + \Delta y \quad \textbf{(6-7a)}$$

or

$$u' = u'_1 + \Delta u' \text{ , } v' = v'_1 + \Delta v' \quad \textbf{(6-7b)}$$

$$CCT = CCT_1 + \Delta\ CCT \quad \textbf{(6-8a)}$$

$$CRI = CRI_1 + \Delta\ CRI \quad \textbf{(6-8b)}$$

$$R_9 = R_{9,1} + \Delta R_9 \quad \textbf{(6-8c)}$$

6.2.2 Measurement at Temperature $T_{b,0}$ + ΔT. A temperature difference, ΔT, is selected or provided. ΔT may be a positive value or a negative value but should not be 25 °C. The temperature controlling method shall be adjusted so that $T_b = T_{b,0} + \Delta T$. This value is recorded as $T_{b,2}$. The electrical power P_2, intensity I_2, total luminous flux Φ_2, radiant flux, or photon flux shall be measured. Optionally, chromaticity coordinates, (x_2, y_2); or (u'_2, v'_2), correlated color temperature CCT_2; and/or SPD may be measured.

Record the corrected results for $T_b = T_{b,2}$ as:

$$P = C_{power} P_2 \quad \textbf{(6-9)}$$

$$\Phi = C_{flux} \Phi_2 \quad \textbf{(6-10a)}$$

or:

$$\Phi = C_{flux} I_2 \quad \textbf{(6-10b)}$$

And optionally:

$$x = x_2 + \Delta x \text{ , } y = y_2 + \Delta y \quad \textbf{(6-11a)}$$

or:

$$u' = u'_2 + \Delta u' \text{ , } v' = v'_2 + \Delta v' \quad \textbf{(6-11b)}$$

$$CCT = CCT_2 + \Delta\ CCT \quad \textbf{(6-12a)}$$

$$CRI = CRI_2 + \Delta\ CRI \quad \textbf{(6-12b)}$$

$$R_9 = R_{9,2} + \Delta R_9 \quad \textbf{(6-12c)}$$

6.2.3 Measurement at Additional Temperatures. Additional measurements at other temperature other than $T_{b,0}$ + 25 °C and $T_{b,0} + \Delta T$ may be carried with the procedure in **Section 6.2.1**. The calculated results shall be in a similar manner to that given in **Section 6.2.2**.

7.0 Reporting Requirements

The following shall be included in the test report for each T_b at each condition:

- Test date
- Test facility
- Test instrumentation
- DUT description
- Internal procedure reference
- Input power (W)
- Input voltage (V)
- Input current (A)O
- Luminous flux (lm), radiant flux, or photon flux
- Luminous efficacy (lm/W), radiant efficiency, or photon efficacy (micromol/J)
- Spectral Power Distribution (SPD) (optional)
- Chromaticity [(x, y) or (u′, v′)] (optional)
- Correlated color temperature (K) (optional)
- T_b or T_d
- Uncertainty statement (optional)

Table 7-1 shows the recommended table format for the test report.

Table 7-1. Recommended Table Format for the Test Report

Test date, facility, equipment, and operator:			
DUT description (manufacturer, description, catalog number, etc.):			
If applicable, DUT driver description (manufacturer, description, catalog number, and input and output parameters):			
Description of test method including testing configuration:			
Internal procedure reference:			
	Ambient Temperature (Initial)	**First Elevated Temperature (Initial + 25 °C)**	**Second Elevated Temperature (per Test Requestors)**
Measured temperature of T_b (or T_d)			
Input power (W)			
Input voltage (V)			
Input current (A)			
Luminous flux (lm), radiant flux, or photon flux			
Luminous efficacy (lm/W), radiant efficiency, or photon efficacy			
Spectral power distribution (SPD)			
CIE chromaticity [(x, y) or (u', v')] (optional)			
Correlated color temperature (K) (optional)			
Spectral power distribution (SPD)			
Uncertainties			

8.0 INFORMATIVE REFERENCES

ASSIST Recommends. Recommendation for testing and evaluating white LED light engines and integrated LED lamps used in decorative lighting luminaires. Troy, NY: Lighting Research Center, Rensselaer Polytechnic Institute. ASSIST Recommends. 2008 May(R2009 Apr);4(1).

ASTM International. ASTM Standard E230-03, Standard Specification and Temperature-Electromotive Force (EMF) Tables for Standardized Thermocouples. West Conshohocken, Penn.: ASTM International; 2003.

National Bureau of Standards. Experimental Statistics – National Bureau of Standards Handbook 91, Chapter 1. Washington, DC: U.S. Government Printing Office; 1963.

Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

a) Proposed change accepted for public review without modification
b) Proposed change accepted for public review with modification
c) Proposed change accepted for further study
d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

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